Geometry : Vectors — problem 12.1.36

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The convex quadrilateral $ABCD$ is cut by its diagonals (intersecting in O) into four triangles AOB, BOC, COD, DOA. Let S_1 and S_2 be the centroids of the first and third of those triangles, and H_1, H_2 the orthocenters of the other two triangles. Then $H_1H_2 \perp S_1S_2$.